

NutriX: Vision-Grounded Nutrition Understanding and Constrained Meal Planning for Culturally Diverse Foods

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Abstract

Food recognition from images is incorporated into automated dietary assessment, but it is critical for real nutrition systems to link visual understanding of food with knowledge about the various food types across cultures and dietary planning. In this paper, we present NutriX, a vision-based nutrition framework that links food detection to goal-based meal planning with nutrition grounding while separating visual recognition from food knowledge. The vision component uses a YOLOv8n object detection model trained on 123 food categories with 56,147 images and achieves 58.78% mAP@50 and 40.56% mAP@50:95. The additional barcode and Optical Character Recognition (OCR) modalities are included to provide further identification cues for packaged food and nutrition labels. NutriX also integrates a food knowledge-based layer with 217 classes and 10 cuisines providing the link to recipes, foods, ingredients, and nutrition facts. Instead of treating these 217 classes as a visual benchmark, the layer provides knowledge to identify culturally diverse foods and link them to nutrition and recipes. The vision part shows 81.56% mAP@50 on an Indian food dataset and results of semantic label normalization and orientation evaluation are reported as additional contributions of this paper. The final framework provides a useful way of translating visual food comprehension into culturally informed nutrition knowledge and constrained meal planning.

1. Introduction

Food recognition from images is an important component of automated dietary assessment, but visual recognition alone is insufficient for practical nutrition systems. A useful dietary system must connect detected food items with nutritional information, dietary requirements, recipes, and meal-planning decisions. This creates a systems-level challenge that spans computer vision, food identification, nutrition knowledge, and constrained meal planning.

The challenge becomes more pronounced for culturally diverse foods. Visual recognition datasets typically organize foods into predefined image-based categories, whereas nutrition and food knowledge resources represent foods through ingredients, recipes, nutritional attributes, dietary properties, and regional terminology. Consequently, the semantic granularity of a visual taxonomy does not necessarily match that of a broader food-knowledge resource. Visually similar foods may also have different identities, while the same food concept may appear under multiple names across cuisines. Treating every knowledge-base entry as a visual recognition class therefore conflates two distinct functions: recognizing food from an image and representing food within a broader nutrition knowledge space.

NutriX addresses this gap by separating visual recognition from broader food knowledge while connecting the two through food identity normalization. The framework uses a YOLOv8n-based detector for 123 visual food classes and complements visual evidence with barcode and optical character recognition pathways for packaged foods and nutrition labels. Detected food identities are normalized and linked to structured nutritional and recipe information. A separate regional food-knowledge layer contains 217 food classes spanning 10 cuisines, together with associated ingredients, recipes, nutritional attributes, and dietary information. This layer is not treated as a 217-class visual recognition benchmark; instead, it provides a broader knowledge space through which recognized foods can be grounded for culturally diverse nutrition applications. The resulting representations are then used by a constrained meal-planning component incorporating nutritional targets and user-specific requirements.

We evaluate NutriX across multiple levels of the system. The visual component is evaluated using standard object-detection metrics and complementary analyses designed to examine class representation, culturally relevant food subsets, semantic label ambiguity, rotational robustness, and qualitative failure modes. The downstream meal-planning component is evaluated independently using con-

strained optimization instances over the recipe knowledge base. Together, these evaluations examine not only visual recognition performance but also how taxonomy design, cultural food representation, and downstream constraints affect the use of food-vision systems in nutrition-oriented applications.

The main contributions of this work are:

- A vision-grounded nutrition framework that connects food detection, multimodal identification, food identity normalization, nutrition and regional knowledge grounding, and constrained meal planning while maintaining a distinction between visual recognition and broader food representation.
- A 123-class YOLOv8n food detector trained on 56,147 annotated images, integrated with complementary barcode and OCR identification pathways and a 217-class regional food-knowledge layer spanning 10 cuisines.
- A multi-level evaluation of the framework, including overall food-detection performance, comparison with a culturally relevant Indian-food subset, semantic label normalization, rotational robustness, qualitative failure analysis, and downstream meal-planning performance across different candidate-pool sizes.

The remainder of the paper describes the NutriX framework, including its visual recognition and multimodal identification pathways, nutrition and regional food-knowledge grounding, and constrained meal planning. We then present the experimental setup and results, followed by a discussion of the findings and limitations.

2. Related Work

2.1. Food Recognition and Dietary Assessment

Food recognition has been studied through image classification, fine-grained recognition, and object detection. Food-101 established a widely used benchmark containing 101 food categories and 101,000 images, providing a foundation for large-scale visual food recognition [2]. More recent resources have extended food recognition toward nutrition-oriented applications, including AI4Food-NutritionDB, which combines food recognition with structured nutritional information [15]. Nutrition5k further connects visual food understanding with food weight and nutritional attributes, demonstrating the value of jointly modeling visual and nutritional information [16].

Object detection is particularly relevant to real-world food images because multiple food items may appear within the same scene. Detection-based approaches therefore provide localized food identities that can be connected to downstream semantic and nutritional resources. NutriX builds its visual recognition component using YOLOv8n [8], while addressing the additional challenge of connecting detected food identities to structured food

knowledge.

2.2. Multimodal Food Identification

Visual recognition alone may be insufficient for packaged foods or cases where identifying information is primarily available through product codes or text. Product-oriented resources such as Open Food Facts provide structured information that can complement visual food identification [11]. Multimodal nutrition systems have also combined visual recognition with nutritional analysis and personalized recommendations [5, 18].

These approaches motivate the use of complementary identification signals rather than relying exclusively on visual evidence. NutriX therefore incorporates barcode scanning and optical character recognition (OCR) alongside visual food detection, allowing food identities to be established from complementary sources when visual evidence is insufficient or ambiguous.

2.3. Food and Nutrition Knowledge

Structured food and nutrition knowledge provides the semantic layer required to translate recognized food identities into nutritional and dietary information. Nutrition databases provide quantitative attributes, while recipe resources additionally capture ingredients, preparation information, serving information, and dietary properties. Knowledge-oriented approaches have explored the representation and use of food concepts for nutrition and health applications [3, 10].

Downstream nutrition systems can use these resources for ranking, recommendation, or constrained meal planning. Existing approaches have considered nutritional targets, dietary preferences, and other user requirements when generating meal recommendations [7, 12]. Other work has incorporated cultural food considerations into nutrition-oriented systems [9]. These studies demonstrate the importance of structured food knowledge and downstream decision-making, but the visual recognition taxonomy and broader food-knowledge taxonomy are not necessarily aligned.

2.4. Positioning of NutriX

Table 1 provides a capability-level comparison of representative food-recognition, nutrition, and meal-planning systems. The comparison is qualitative because the referenced systems address different tasks and use different datasets and evaluation protocols.

NutriX is positioned at the intersection of these research directions by connecting visual food recognition with multimodal identification, structured nutrition and recipe knowledge, culturally diverse food representations, and constrained meal planning. A key distinction is that the visual recognition taxonomy and regional food-knowledge taxon-

Table 1. Capability-level comparison of representative food-recognition, nutrition, and meal-planning systems.

Work	Vision	Nutrition	Personal.	Cultural	Planning	Optim.
Food-101 [2]	✓	–	–	–	–	–
Nutrition5k [16]	✓	✓	–	–	–	–
AI4Food-NutritionDB [15]	✓	✓	–	–	–	–
Chen et al. [3]	–	✓	✓	–	–	–
Li et al. [10]	–	✓	✓	–	–	–
Islam et al. [7]	–	✓	✓	–	✓	✓
NutriAI [5]	✓	✓	✓	–	✓	–
Papastratis et al. [12]	–	✓	✓	–	✓	✓
Kalpakoglou et al. [9]	–	✓	✓	✓	✓	–
NutriX	✓	✓	✓	✓	✓	✓

omy operate at different semantic granularities: NutriX uses 123 visual food classes while maintaining a broader 217-class regional knowledge layer spanning 10 cuisines. The regional layer is therefore used for semantic, nutritional, and recipe grounding rather than as an additional visual recognition benchmark.

This system-level integration enables recognized food identities to be normalized and linked to nutritional and recipe information before being provided to a constrained meal-planning component. The resulting framework connects food-vision research with downstream nutrition-oriented decision making while explicitly accounting for heterogeneous food labels and culturally diverse food knowledge.

3. NutriX Framework

NutriX is a vision-grounded nutrition framework that converts food observations from visual and complementary identification signals into canonical food identities, grounds these identities in structured nutrition and regional food knowledge, and uses the resulting information for constrained meal planning. An overview of the complete pipeline is shown in Fig. 1.

3.1. System Overview

NutriX accepts food information through multiple complementary pathways. Food images are processed by the visual detection component, while packaged products and text-based nutrition information can additionally be identified through barcode scanning and optical character recognition (OCR). The resulting evidence is mapped to a canonical food representation that serves as the interface between food identification and downstream nutrition processing.

The canonical representation associates a food identity with structured attributes such as category, ingredients, food type, nutritional values, recipes, serving information, and dietary properties. This representation allows different identification pathways to provide a consistent input to the same downstream nutrition and meal-planning components.

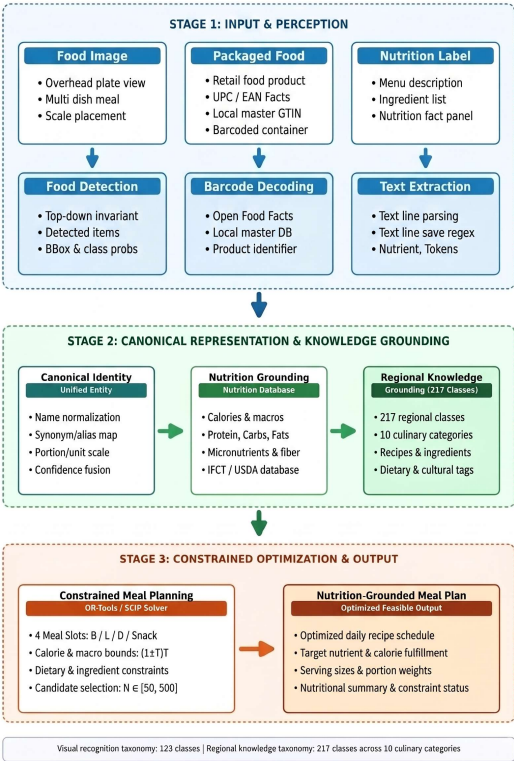


Figure 1. Overview of the NutriX vision-grounded nutrition pipeline. Food images and complementary identification signals are converted into canonical food identities, grounded in structured nutrition and regional food knowledge, and passed to constrained meal planning.

3.2. Vision-Based Food Detection

The primary visual component uses a YOLOv8n object detector [8]. The detector operates on a 123-class food taxonomy and is trained using 56,147 annotated images. As an object detector, it can identify multiple food items and their locations within a single image, making it suitable for meal and plate images containing multiple food regions.

For each detected object, the predicted food category and spatial location are retained. The predicted category is subsequently passed to the food normalization stage, where it is mapped to the corresponding canonical food identity. Training configuration, dataset composition, and augmentation settings are described in Section 4.

3.3. Multimodal Food Identification

Visual recognition is complemented by barcode and optical character recognition (OCR) pathways. Barcode identification provides product-level evidence for packaged foods, while OCR extracts food names and nutrition-related information from textual labels.

These pathways are treated as complementary sources of

evidence rather than independent systems. Their outputs are converted into the same canonical food representation used by the visual pathway, allowing the downstream nutrition-processing components to operate on a consistent food identity when visual evidence is unavailable, ambiguous, or insufficient.

3.4. Nutrition Knowledge Grounding

Once a canonical food identity is established, NutriX links it to structured nutrition and food knowledge. The nutrition layer contains nutrient values together with associated recipes, ingredients, serving information, and dietary attributes. The framework incorporates structured food-composition resources including the Indian Food Composition Tables (IFCT 2017) and USDA FoodData Central [6, 17].

This grounding step converts a recognized food identity into structured information that can be used both for nutrition analysis and for downstream recipe selection.

3.5. Regional Food Knowledge

NutriX maintains a separate regional food-knowledge layer containing 217 food classes spanning 10 cuisines, together with associated recipes, ingredients, nutritional attributes, serving information, and dietary properties. The regional layer operates as a knowledge resource rather than as an additional visual recognition taxonomy.

The distinction between the two layers is intentional. The visual component operates on 123 food classes for which image-based detection is defined, whereas the broader regional layer provides additional food concepts and contextual information that can be used for semantic and nutritional grounding. This separation allows the system to represent culturally diverse foods without requiring every knowledge-base entry to correspond to a visual detector class.

3.6. Constrained Meal Planning

The grounded food and nutrition representations are passed to a constrained meal-planning component implemented using Google OR-Tools with the SCIP solver [4]. The planner selects recipes for predefined meal slots while considering nutritional targets and user-specific requirements.

Let $\mathcal{R}$ denote the candidate recipe set and $\mathcal{M}$ the set of meal slots. For nutritional attribute n , the aggregate daily quantity is

$$N_n = \sum_{m \in \mathcal{M}} \sum_{r \in \mathcal{R}} q_{m,r} v_{r,n}, \quad (1)$$

where $q_{m,r}$ denotes the selected quantity of recipe r in meal slot m , and $v_{r,n}$ denotes the contribution of recipe r to nutritional attribute n .

Table 2. Composition of the NutriX visual food-recognition dataset.

Split	Images	Classes	Purpose
Train	51,535	123	Model training
Validation	2,484	123	Model selection
Test	2,128	123	Final evaluation
Total	56,147	123	–

The planner uses binary recipe-selection variables to assign recipes to meal slots while enforcing meal-slot, nutritional, and user-specific constraints. Nutritional requirements include target ranges for calories and macronutrients, while additional restrictions can encode dietary preferences, allergies, and other meal requirements. Recipe repetition across meal slots can also be restricted to encourage meal diversity.

When an instance is infeasible under the initial nutritional tolerance, NutriX supports adaptive relaxation of the nutritional bounds before determining whether a feasible meal plan can be produced. The optimization process therefore provides a deterministic mapping from grounded food and nutrition information to a constrained set of meal combinations.

4. Experimental Setup

We evaluate NutriX at two levels: visual food recognition and downstream meal planning. The visual evaluation examines overall detection performance, class representation, culturally relevant food subsets, semantic label ambiguity, rotational robustness, and qualitative failure modes. The meal-planning component is evaluated independently using constrained optimization instances constructed from the recipe knowledge base.

4.1. Visual Dataset and Training

The NutriX visual food-recognition dataset contains 56,147 annotated images spanning 123 food classes. The images were collected from multiple food-image sources covering fruits and vegetables, cooking ingredients, and Indian food categories [1, 13, 14]. The dataset contains variation in food appearance, viewpoint, background, scale, and lighting to reflect practical food-photography conditions.

The dataset is divided into independent training, validation, and test splits as summarized in Table 2.

The visual recognition component uses YOLOv8n initialized from the pretrained `yolov8n.pt` checkpoint [8]. The model was trained for 30 epochs with a batch size of 16 and an input resolution of 640×640 pixels. The optimizer was configured as `auto`, with the recorded learning-rate and weight-decay settings retained.

Spatial augmentation was used to improve robustness to

variation in food orientation and appearance. The training configuration included rotation up to 180° , scale variation of 0.2, translation of 0.1, horizontal flipping with probability 0.5, and mosaic augmentation. HSV-based color perturbation was also enabled, while mixup and perspective augmentation were not used. These augmentation settings form part of the final training configuration and were not evaluated through a separate augmentation ablation.

4.2. Evaluation Protocol

The visual component is evaluated using precision, recall, mAP@50, and mAP@50:95. The reported metrics follow standard object-detection evaluation procedures, with mAP@50 computed at an IoU threshold of 0.50 and mAP@50:95 averaged over IoU thresholds from 0.50 to 0.95.

We additionally analyze the distribution of food instances across the 123 classes to assess class representation imbalance and its implications for interpreting aggregate performance.

To examine culturally relevant performance, the trained detector is evaluated on a subset containing 31 Indian food classes. Precision, recall, mAP@50, and mAP@50:95 are reported for comparison with the complete 123-class evaluation.

Food datasets compiled from multiple sources may contain labels representing the same or closely related food concepts. We therefore perform a post-hoc semantic normalization analysis on two groups, *brinjal/eggplant* and *capsicum/bell pepper*. The existing trained detector is used without retraining, and predictions are evaluated first using the original labels and then after grouping the semantically equivalent labels.

Rotational robustness is evaluated using a fixed subset of 200 validation images at four orientations: 0° , 90° , 180° , and 270° . Ground-truth bounding boxes are transformed consistently with each rotated image. The same trained checkpoint is evaluated under all four conditions using precision, recall, and mAP@50. This experiment measures robustness to image orientation rather than the causal contribution of rotational augmentation.

Finally, representative validation predictions are examined qualitatively to characterize practical failure modes, including background false positives, missed detections, visually similar foods, semantic-label confusion, crowded multi-object scenes, and low-confidence detections.

4.3. Regional Food Knowledge

The regional food-knowledge layer is evaluated independently of the visual dataset. It contains 217 food classes spanning 10 cuisines, together with associated recipes, ingredients, nutritional attributes, serving information, and dietary properties. Because the regional repository does not

provide complete image coverage for all 217 classes, it is not evaluated as a 217-class visual recognition benchmark.

For analyses involving visual recognition, only regional food classes with an established correspondence to the 123-class visual taxonomy are considered. This allows culturally relevant subsets to be analyzed using the existing visual detector while preserving the broader regional layer as a knowledge resource.

4.4. Meal-Planning Benchmark

The downstream meal-planning component is evaluated independently from visual detection using 80 optimization instances constructed from a recipe knowledge base containing 9,139 recipes. Twenty instances are evaluated for each candidate-pool size $N \in \{50, 100, 300, 500\}$. Each optimization instance considers four meal slots: breakfast, lunch, dinner, and snack.

The OR-Tools/SCIP solver selects recipes subject to nutritional and user-specific constraints. We report feasibility, mean solve time, calorie error, and protein error for each candidate-pool size. When an instance is infeasible under the initial nutritional tolerance, adaptive relaxation is applied beginning at $\tau = 0.15$, increasing in increments of 0.05 up to a maximum tolerance of $\tau = 0.40$. The effect of this relaxation on overall feasibility is reported in Section 5.

5. Results

We evaluate NutriX through a series of comparative analyses spanning visual recognition and downstream meal planning. These comparisons examine overall detection performance against the culturally relevant Indian-food subset, the effect of semantic label normalization, robustness across image orientations, and meal-planning behavior across different candidate-pool sizes.

5.1. Overall Food Detection Performance

The final YOLOv8n checkpoint achieves a precision of 71.08%, recall of 51.32%, mAP@50 of 58.78%, and mAP@50:95 of 40.56% on the 123-class test set (Table 3). The best validation mAP@50 observed during training was 59.32% at epoch 27, while the final checkpoint at epoch 30 provides the reported evaluation results.

The difference between precision and recall indicates that missed detections remain a substantial source of error for the visual component. The lower mAP@50:95 relative to mAP@50 also indicates that localization becomes more challenging under stricter IoU thresholds.

5.2. Class Distribution and Evaluation Coverage

The 123 food classes are unevenly represented across the dataset splits (Table 4). The validation split contains 2,484

Table 3. Overall performance of the NutriX food detector on the 123-class test set.

Metric	Score
Precision	71.08%
Recall	51.32%
mAP@50	58.78%
mAP@50:95	40.56%

Table 4. Class-associated image-count statistics across the NutriX dataset splits. Counts are computed per food class; therefore, an image containing multiple classes contributes to multiple class counts.

Split	Min	Max	Mean	Median
Train	0	5,014	767.95	165
Validation	0	164	21.41	3
Test	0	141	19.44	7

Table 5. Comparison between the complete visual benchmark and the Indian food subset.

Metric	123-Class	Indian Subset
Classes	123	31
Precision	71.08%	78.94%
Recall	51.32%	74.70%
mAP@50	58.78%	81.56%
mAP@50:95	40.56%	63.33%

images and 6,109 identified food instances. The mean number of images per class is 21.41, while the median is only 3. Because an image may contain multiple food classes, class-associated counts are not equivalent to unique image counts.

Thirty-seven classes have no images in the validation split, 30 classes contain fewer than five images, and only 51 classes contain more than 20 images. This uneven representation limits the extent to which aggregate metrics can be interpreted as uniform estimates of performance across all classes. Figure 2(a) illustrates the resulting variation in class representation.

5.3. Indian Food Subset Evaluation

The Indian-food subset provides a direct comparison with the complete 123-class test set, achieving a precision of 78.94%, recall of 74.70%, mAP@50 of 81.56%, and mAP@50:95 of 63.33% (Table 5).

The absolute mAP@50 difference between the two evaluations is 22.78 percentage points. The improvement should be interpreted in the context of class composition and representation rather than as evidence that Indian foods are inherently easier to recognize.

Several Indian food classes achieve high AP@50, including jalebi (99.10%), onion pakoda (99.10%), palak paneer (96.40%), poha (96.50%), and chole (93.40%). Lower-

Table 6. Semantic label normalization for visually overlapping food categories.

Semantic Group	Precision	Recall	F1
Brinjal/Eggplant	92.07%	71.41%	80.44%
Capsicum/Bell Pepper	88.77%	86.27%	87.50%

performing examples include dosa (67.90%), fish curry (63.10%), and shahipaneer (69.40%). Rahar ko daal has an AP@50 of 0% in the available validation samples, further illustrating the effect of limited class representation on class-level performance.

5.4. Semantic Normalization Results

Semantic normalization changes the measured performance for both analyzed label groups (Table 6). For the brinjal/eggplant group, the normalized representation achieves 92.07% precision, 71.41% recall, and an F1 score of 80.44%. For capsicum/bell pepper, the corresponding values are 88.77%, 86.27%, and 87.50%.

For the brinjal/eggplant group, the original *brinjal* category has zero recall, whereas the corresponding *eggplant* category achieves 71.41% recall. Grouping the two labels under a common semantic identity therefore changes the measured performance substantially. The capsicum/bell-pepper group shows a similar effect.

These results indicate that part of the apparent class-level error can originate from taxonomy fragmentation. When semantically equivalent or closely related food names are represented as separate classes, correct recognition of the underlying food concept may nevertheless be counted as a classification error.

5.5. Rotational Robustness

The detector remains stable across the four evaluated orientations (Table 7). mAP@50 decreases from 89.24% at 0° to 89.11% at 90°, 88.75% at 180°, and 88.04% at 270°.

The maximum absolute mAP@50 difference across the four orientations is 1.20 percentage points. Precision and recall likewise remain relatively stable, indicating limited sensitivity to the tested orthogonal rotations.

Because the same trained checkpoint is used for all four conditions, this experiment demonstrates rotational robustness rather than isolating the causal contribution of rota-

Table 7. Rotational robustness of the NutriX detector on 200 validation images.

Rotation	Precision	Recall	mAP@50
0°	87.10%	84.32%	89.24%
90°	86.04%	83.43%	89.11%
180°	88.08%	82.68%	88.75%
270°	87.06%	81.57%	88.04%

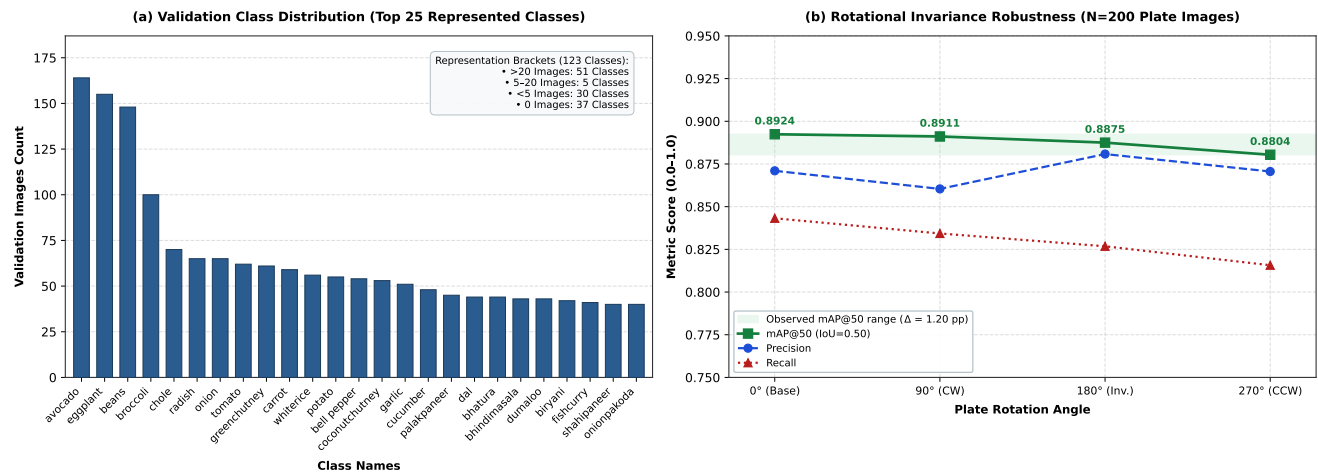


Figure 2. Quantitative evaluation of the NutriX visual component. (a) Validation-set class representation for the 123-class detector, showing the 25 most represented classes together with the overall representation brackets. (b) Rotational robustness on 200 validation plate images evaluated at 0°, 90°, 180°, and 270°.

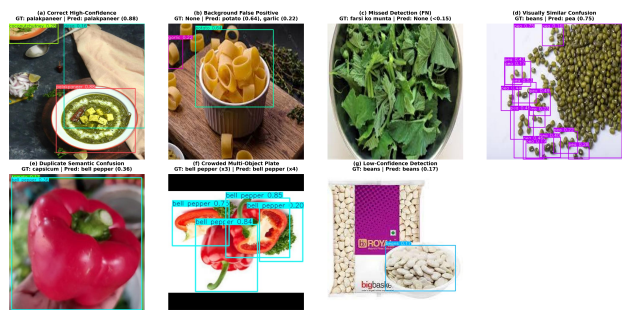


Figure 3. Qualitative food-recognition and error analysis across representative validation examples. The examples illustrate (a) correct high-confidence detection, (b) background false positive, (c) missed detection, (d) visually similar confusion, (e) semantic label confusion, (f) crowded multi-object scenes, and (g) low-confidence detection.

tional augmentation.

5.6. Qualitative Error Analysis

Figure 3 presents representative validation predictions covering correct high-confidence detections, background false positives, missed detections, visually similar foods, semantic-label confusion, crowded multi-object scenes, and low-confidence detections.

The examples reveal several recurring sources of error. Background objects can produce false positives, while visually similar foods can result in incorrect class assignments. Sparse class representation contributes to missed detections, and semantically overlapping labels can produce apparent classification errors despite strong localization. In one representative capsicum/bell-pepper example, the pre-

Table 8. Meal-planning optimization benchmark over 20 instances per candidate-pool size.

Pool	Feas.	Mean (ms)	Cal. Err.	Prot. Err.
50	50.0%	39.25	21.06%	27.70%
100	40.0%	43.86	22.23%	28.56%
300	40.0%	81.47	22.74%	28.41%
500	40.0%	89.23	22.74%	28.41%
Tolerance relaxation				
Initial tolerance				15%
Tolerance increment				5%
Maximum tolerance				40%
Initial feasibility				25.0%
Final feasibility				42.5%

dicted bounding box achieves an IoU of 0.912 despite a mismatch between the ground-truth and predicted class names, illustrating the distinction between localization quality and semantic-label agreement.

5.7. Meal Optimization Benchmark

The meal-planning benchmark compares optimization behavior across candidate-pool sizes of 50, 100, 300, and 500 recipes (Table 8). Mean solve time increases from 39.25 ms for a pool of 50 recipes to 43.86 ms, 81.47 ms, and 89.23 ms for pools of 100, 300, and 500 recipes, respectively.

Feasibility ranges from 40.0% to 50.0% across the four candidate-pool sizes. Calorie error ranges from 21.06% to 22.74%, while protein error ranges from 27.70% to 28.56%.

Adaptive nutritional tolerance relaxation improves overall feasibility from 25.0% at the initial 15% tolerance to 42.5% at the maximum 40% tolerance. The remaining infeasible instances indicate that the simultaneous nutritional, regional, and meal-slot constraints can substantially restrict the feasible solution space.

6. Discussion

The results highlight that the performance of a food-vision system depends not only on the underlying detector but also on how food concepts are represented in the dataset and connected to downstream knowledge. The overall detection results provide a practical baseline for the 123-class visual taxonomy, while the additional analyses reveal several factors that influence how these aggregate metrics should be interpreted.

The higher performance observed on the Indian-food subset is closely associated with the composition and representation of the evaluated classes. The subset contains fewer classes than the complete visual taxonomy and therefore represents a different recognition setting. Its higher performance should not be interpreted as evidence that Indian foods are inherently easier to recognize. Instead, the result demonstrates the importance of evaluating culturally relevant subsets when a food-recognition system is intended for diverse dietary contexts.

The semantic normalization experiment further shows that evaluation outcomes depend on the granularity of the underlying food taxonomy. Labels such as *brinjal/eggplant* and *capsicum/bell pepper* can represent closely related or equivalent food concepts while remaining separate detection classes. As a result, a prediction can be visually well localized but still be counted as a class-level error because of a naming mismatch. This supports the separation between visual recognition and canonical food representation used in NutriX.

The rotational evaluation demonstrates stable detector performance across the tested orientations, with only a small change in mAP@50. This indicates that the trained visual component is relatively insensitive to the tested orthogonal rotations. However, because the same checkpoint was evaluated under all conditions and no controlled augmentation ablation was performed, the experiment does not isolate the contribution of rotational augmentation itself.

The meal-planning results demonstrate the computational feasibility of connecting grounded food information to constrained downstream decision making. Increasing the candidate recipe pool increases optimization time, while the measured nutritional errors remain within a relatively narrow range across the evaluated pool sizes. The adaptive tolerance experiment also shows that relaxing nutritional constraints can recover additional feasible solutions. At the same time, the remaining infeasible instances demonstrate that satisfying multiple nutritional, regional, and meal-slot requirements simultaneously can substantially restrict the feasible solution space.

Several limitations should be considered when interpreting these findings. First, the broader 217-class regional food-knowledge layer does not constitute a complete visual benchmark, and therefore its coverage should not be

interpreted as equivalent to that of the 123-class detector. Second, the visual dataset is unevenly distributed across classes, with some classes having limited representation in the evaluation splits. Third, the rotational experiment evaluates robustness but does not provide a controlled ablation of the training augmentation strategy. Finally, the meal-planning evaluation measures computational feasibility and nutritional constraint satisfaction rather than clinical or behavioral outcomes.

Taken together, the results support NutriX as a vision-grounded nutrition-processing framework that connects heterogeneous food recognition with structured and culturally diverse food knowledge. The findings also indicate that future improvements should address class representation, semantic taxonomy design, broader visual coverage of regional foods, and more systematic evaluation of training and downstream planning components. NutriX should therefore be viewed as a technical framework for nutrition-oriented food understanding and planning rather than as a clinically validated dietary decision system.

7. Conclusion

This work presented NutriX, a vision-grounded nutrition framework that connects food recognition with multimodal identification, structured nutrition knowledge, culturally diverse food representations, and constrained meal planning. The framework separates the 123-class visual recognition taxonomy from the broader 217-class regional food-knowledge layer, allowing visual food identities to be connected to nutritional and recipe information without requiring every knowledge-base entry to correspond to a visual detector class.

The YOLOv8n-based visual component achieved 58.78% mAP@50 and 40.56% mAP@50:95 on the 123-class test set, while the evaluated Indian-food subset achieved 81.56% mAP@50. Additional analyses demonstrated the effects of semantic label normalization and showed stable performance under the tested image rotations. The downstream meal-planning evaluation further demonstrated the computational feasibility of constrained recipe selection under nutritional and user-specific requirements.

Overall, the results show that practical food-vision systems benefit from separating visual recognition from broader food knowledge and explicitly accounting for taxonomy, cultural representation, and downstream constraints. NutriX provides a unified framework for connecting these components and establishes a foundation for further work on broader visual food coverage, improved semantic representations, and more comprehensive nutrition-oriented evaluation.

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